

Chapter 1



Data Analysis

Section 1.1 Analyzing Categorical Data



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Data Analysis

LEARNING TARGETS

By the end of this section, you should be able to:

- ✓ MAKE and INTERPRET bar graphs for categorical data.
- ✓ IDENTIFY what makes some graphs of categorical data misleading.
- ✓ CALCULATE marginal and joint relative frequencies from a two-way table.
- ✓ CALCULATE conditional relative frequencies from a two-way table.
- ✓ Use bar graphs to COMPARE distributions of categorical data.
- ✓ DESCRIBE the nature of the association between two categorical variables.



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Organizing Categorical Data

Province	Gender	Number of languages spoken	Handedness	Height (cm)	Wrist circumference (mm)	Preferred communication
Saskatchewan	Male	1	Right	175.0	180	In person
Ontario	Female	1	Right	162.5	160	In person
Alberta	Male	1	Right	178.0	174	Facebook
Ontario	Male	2	Right	169.0	160	Cell phone
Ontario	Female	2	Right	166.0	65	In person
Nunavut	Male	1	Right	168.5	160	Text messaging
Ontario	Female	1	Right	166.0	165	Cell phone
Ontario	Male	4	Left	157.5	147	Text messaging
Ontario	Female	2	Right	150.5	187	Text messaging
Ontario	Female	1	Right	171.0	180	Text messaging

Categorical variable

Values
(These are the data)

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Organizing Categorical Data

Frequency table		Relative frequency table	
Preferred method	Frequency	Preferred method	Relative frequency
Cell phone	2	Cell phone	$2/10 = 0.20$ or 20%
Facebook	1	Facebook	$1/10 = 0.10$ or 10%
In person	3	In person	$3/10 = 0.30$ or 30%
Text messaging	4	Text messaging	$4/10 = 0.40$ or 40%

Count

Proportion

Percent



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Displaying Categorical Data

To display the distribution of categorical data, make a **bar graph**

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Text messaging	4	Text messaging	$4/10 = 0.40$ or 40%

Count

Proportion

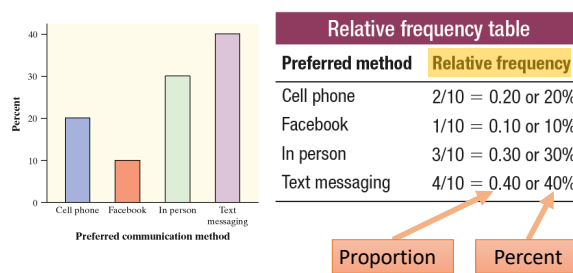
Percent

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Displaying Categorical Data

To display the distribution of categorical data, make a **bar graph**



Proportion

Percent

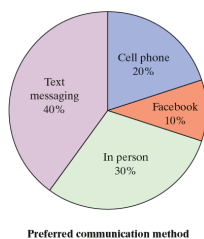
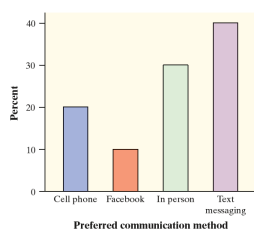


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Displaying Categorical Data

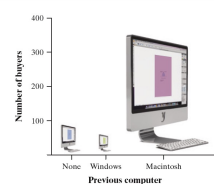
To display the distribution of categorical data, make a **bar graph** or a **pie chart**.



Graphs: Good and Bad

Bar graphs are a bit dull to look at. It is tempting to replace the bars with pictures or to use special 3-D effects to make the graphs seem more interesting.

Don't do it!



CAUTION:
1) beware the pictograph
2) watch those scales

Analyzing Data on Two Categorical Variables

How do you analyze data do when a data set involves two categorical variables?

Respondent	Environmental club?	Snowmobile use
1	No	Own
2	No	Rent
3	Yes	Never
4	Yes	Rent
5	No	Never
...	...	...

A **two-way table** is a table of counts that summarizes data on the relationship between two categorical variables for some group of individuals.

We can include row and column totals

Snowmobile use	Environmental club		Total
	No	Yes	
Never used	445	212	657
Snowmobile renter	497	77	574
Snowmobile owner	279	16	295
Total	1221	305	1526

Snowmobile use	Environmental club member?	
	No	Yes
Never	445	212
Rent	497	77
Own	279	16

Analyzing Data on Two Categorical Variables

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A **marginal relative frequency** gives the percent or proportion of individuals that have a specific value for one categorical variable.

Analyzing Data on Two Categorical Variables

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$$\text{No: } \frac{1221}{1526} = 0.800 \text{ or } 80.0\%$$

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No: $\frac{1221}{1526} = 0.800$ or 80.0% Yes: $\frac{305}{1526} = 0.200$ or 20.0%

Never: $\frac{657}{1526} = 0.431$ or 43.1% Rent: $\frac{574}{1526} = 0.376$ or 37.6% Own: $\frac{295}{1526} = 0.193$ or 19.3%

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A marginal relative frequency tells you about only *one* of the variables in a two-way table.

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A **joint relative frequency** gives the percent or proportion of individuals that have a specific value for one categorical variable and a specific value for another categorical variable.

Analyzing Data on Two Categorical Variables

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A **joint relative frequency** gives the percent or proportion of individuals that have a specific value for one categorical variable and a specific value for another categorical variable.

A joint relative frequency helps answer questions involving *both* of the variables in a two-way table.

Analyzing Data on Two Categorical Variables

	Environmental club		
	No	Yes	Total
Never used	445	212	657
Snowmobile renter	497	77	574
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Total	1221	305	1526

What percent of people in the sample are environmental club members **and** own snowmobiles?

A **joint relative frequency** gives the percent or proportion of individuals that have a specific value for one categorical variable and a specific value for another categorical variable.

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What percent of people in the sample are environmental club members **and** own snowmobiles?

$$\frac{16}{1526} = 0.010 = 1.0\%$$

What proportion of people in the sample are not environmental club members **and** never use snowmobiles?

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$$\frac{16}{1526} = 0.010 = 1.0\%$$

What proportion of people in the sample are not environmental club members **and** never use snowmobiles?

$$\frac{445}{1526} = 0.292$$

A **joint relative frequency** gives the percent or proportion of individuals that have a specific value for one categorical variable and a specific value for another categorical variable.

A joint relative frequency helps answer questions involving **both** of the variables in a two-way table.

Relationships Between Two Categorical Variables

Marginal and joint relative frequencies do not tell us much about the **relationship** between environmental club membership and snowmobile use for the people in the sample.

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A **conditional relative frequency** gives the percent or proportion of individuals that have a specific value for one categorical variable among individuals who share the same value of another categorical variable (the condition).

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What percent of environmental club members in the sample are snowmobile owners?

$$\frac{16}{305} = 0.052 = 5.2\%$$

Relationships Between Two Categorical Variables

	Environmental club		Total
	No	Yes	
Never used	445	212	657
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Total	1221	305	1526

The distribution of snowmobile use among environmental club members is called the **conditional distribution** of snowmobile use among environmental club members.

Relationships Between Two Categorical Variables

	Environmental club		Total
	No	Yes	
Never used	445	212	657
Snowmobile renter	497	77	574
Snowmobile owner	279	16	295
Total	1221	305	1526

$$\text{Never: } \frac{212}{305} = 0.695 \text{ or } 69.5\%$$

$$\text{Rent: } \frac{77}{305} = 0.252 \text{ or } 25.2\%$$

$$\text{Own: } \frac{16}{305} = 0.052 \text{ or } 5.2\%$$

The distribution of snowmobile use among environmental club members is called the **conditional distribution** of snowmobile use among environmental club members.

Relationships Between Two Categorical Variables

	Environmental club		Total
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Snowmobile renter	497	77	574
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Total	1221	305	1526

We can find the distribution of snowmobile use among the survey respondents who are not environmental club members in a similar way.

Snowmobile use	Not environmental club members	Environmental club members
Never	$\frac{445}{1221} = 0.364$ or 36.4%	$\frac{212}{305} = 0.695$ or 69.5%
Rent	$\frac{497}{1221} = 0.407$ or 40.7%	$\frac{77}{305} = 0.252$ or 25.2%
Own	$\frac{279}{1221} = 0.229$ or 22.9%	$\frac{16}{305} = 0.052$ or 5.2%

Relationships Between Two Categorical Variables

AP® Exam Tip

- ✓ When comparing groups of different sizes, be sure to use relative frequencies (percents or proportions) instead of frequencies (counts) when analyzing categorical data.
- ✓ Make sure to avoid statements like “More club members never use snowmobiles” when you mean “A greater percentage of club members never use snowmobiles.”

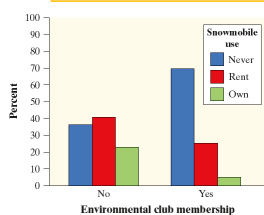
Relationships Between Two Categorical Variables

A **side-by-side bar graph** displays the distribution of a categorical variable for each value of another categorical variable. The bars are grouped together based on the values of one of the categorical variables and placed side by side.

A **segmented bar graph** displays the distribution of a categorical variable as segments of a rectangle, with the area of each segment proportional to the percent of individuals in the corresponding category.

Relationships Between Two Categorical Variables

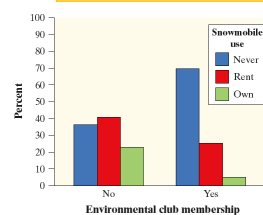
Side-by-side Bar Graph



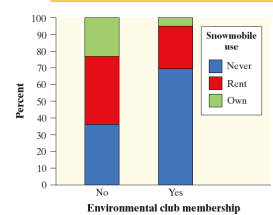
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Relationships Between Two Categorical Variables

Side-by-side Bar Graph



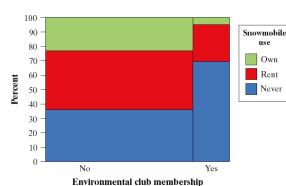
Segmented Bar Graph



Relationships Between Two Categorical Variables

A **mosaic plot** is a modified segmented bar graph in which the width of each rectangle is proportional to the number of individuals in the corresponding category.

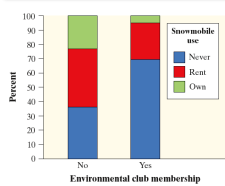
Mosaic Plot



Relationships Between Two Categorical Variables

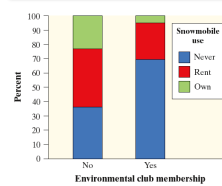
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Relationships Between Two Categorical Variables



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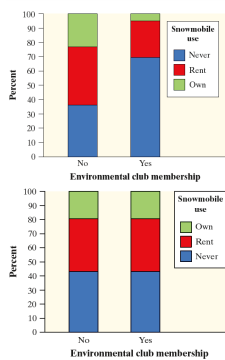
Relationships Between Two Categorical Variables



There is an **association** between two variables if knowing the value of one variable helps us predict the value of the other.

If knowing the value of one variable does not help us predict the value of the other, then there is **no association** between the variables.

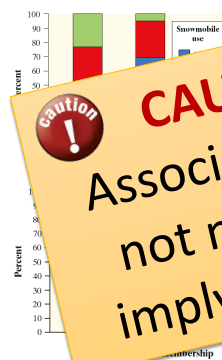
Relationships Between Two Categorical Variables



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Relationships Between Two Categorical Variables



CAUTION:
Association does not necessarily imply causation!

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Section Summary

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- ✓ IDENTIFY what makes some graphs of categorical data misleading.
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- ✓ CALCULATE conditional relative frequencies from a two-way table.
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